KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2016
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

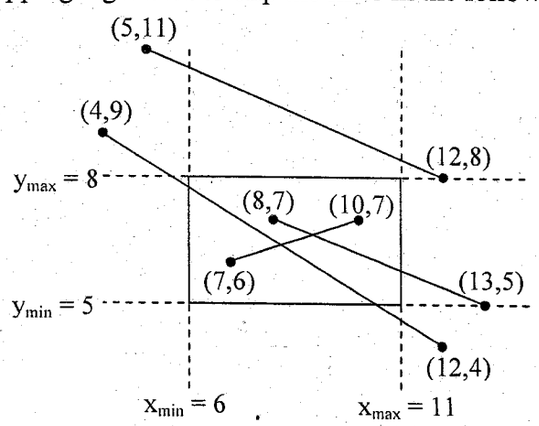
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What does computer graphics deal with? We have cameras to shoot photographs or videos, then what are the driving factors of computer graphics? (07)
   b) Let a unit square defined by $0 \le x \le 1$, and $0 \le y \le 1$ in object coordinate system. New positions X′ are derived from the local positions X by X′ = MX, where M is a transformation matrix. Give a matrix M, expressed in terms of basic modeling transformations, to obtain a rectangle with width = 2 and height = 3, centered around the origin of world coordinate system, and rotated over an angle of 30°. (13)
   c) Why are matrix representations used to describe point transformations in computer graphics? Derive the basic rotation matrix for a rotational transformation about z-axis. (10)
   d) Explain the properties of rigid body transformation and affine transformation. (05)

2. a) Find the mirror reflection of the point (3, 1, 6) with respect to the plane defined by the points (0, 1, 0), (0, 0, 1), and (1, 0, 0). (14)
   b) Explain the working principle of 2D Sutherland-Hodgeman polygon clipping algorithm with necessary illustrations and equations. How can it be extended to 3D? (13)
   c) Briefly explain the taxonomy of planar geometric projection. (08)

3. a) Let the view coordinate system (VCS) is defined by the parameter *eye* $(0, 0, 1)_{WC}$, *look-at* $(0, 0, 0)_{WC}$, *head-up* $\overline{V} = j$. If the projection plane is parallel to the xy plane of the VCS and cuts the negative z-axis of VCS at a distance $d = 1$, find the perspective projection of the point $(1, 1, -1)_{WC}$ on the projection plane considering the VCS origin as a center of projection. (14)
   b) Use Liang-Barsky line clipping algorithm to clip the lines in the following figure. (14)



   c) Briefly explain principal vanishing points for standard perspective projection. Use necessary diagram. (07)

4. a) Let a triangular facet, tessellated part of a large curved surface, is defined by the vertices A (1, 4, 0), B (5, 6, 0), C (4, 0, 0) and their vertex normal are $N_A = -i + j + 3k$, $N_B = j + 3k$, $N_C = i + 4k$ respectively. If the triangle is illuminated by a monochromatic point light source positioned at (–2, 3, 2), and the diffuse reflectivity of the surface is 0.7 then find the diffuse reflection intensity of point P (3, 3, 0) that a viewer will see from position (6, 3, 1). Assume any missing values (if necessary). (15)
   b) Compare among flat shading, Gouraud shading, and Phong shading techniques. (09)
   c) Show that the path of a refracted ray is $\boldsymbol{t} = z - \left(\sqrt{1-|z|^2}\right)\boldsymbol{n}$, where $z = \frac{\eta_1}{\eta_2}(\boldsymbol{u} - (\boldsymbol{u}.\boldsymbol{n}))\boldsymbol{n}$, $\boldsymbol{u}$ is incoming ray, $\boldsymbol{n}$ is surface normal, and η_1, η_2 are the refractive indices of the media. (07)
   d) What are the trade-offs of ray tracing method? (04)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Indicate which raster locations would be chosen by Bresenham's algorithm when scan-converting a line from pixel coordinate (1, 1) to pixel coordinate (8, 5). (12)
   b) What are the major side effects of scan conversion? (08)
   c) Using midpoint algorithm, draw an ellipse with center (2, 2), major axis = 12, and minor axis = 10. (15)

6. a) How do you convert RGB color space into HSI color space? Explain briefly with appropriate figure. (12)
   b) What are the possible representations of a curve? Why is the parametric curve of degree 3 (cubic) preferable? (10)
   c) Define chromatic light and achromatic light. What are the properties of chromatic light? Explain briefly. (07)
   d) What is a raster display? Explain the necessity of scan conversion. (06)

7. a) Define fractal. Generate a fractal image using the following language (12)

   AA[A[B]AA(B)]AAAA(A[B]AA(B))

   where Alphabet {A, B, [,], (,)}

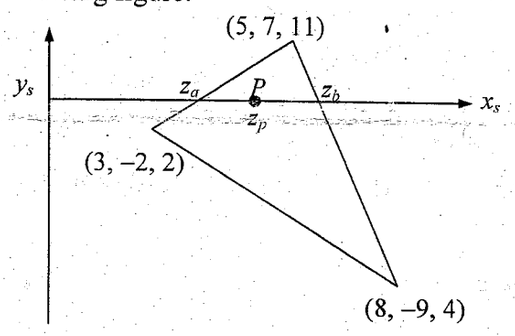
   Rule A ⇒ AA

   B ⇒ A[B]AA(B)

   [] ⇒ left branch

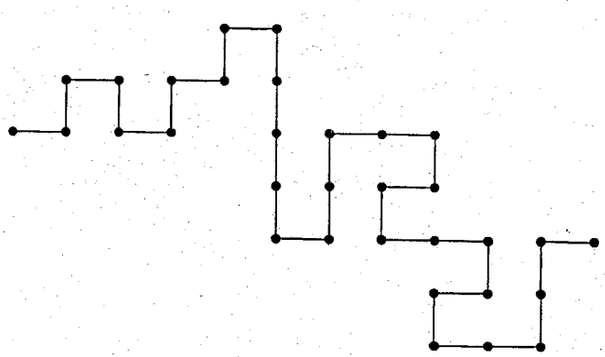
   () ⇒ right branch

   b) What are the difficulties encountered in implementing the painter's algorithm? When does *z*-fighting problem arise? (08)
   c) How does the *z*-buffer algorithm determine which surfaces are hidden and hence compute the z-value at point P of the following figure. (10)



   d) How can you generate the terrain in computer graphics? Explain. (05)

8. a) What do you know about geometric continuity and parametric continuity? Explain G^0, G^1, and G^2 geometric continuity. (10)
   b) Define Hermite surface. Find the geometry vector G_{H_X} for Hermite surface. Also find twist of the vector. (11)
   c) Derive the equation for calculating fractal dimension and hence find the dimension of the following fractals. (07)



   d) What is texturing? Compare between bump mapping and frame mapping techniques. (07)